

CDP Workshop

Composing with Algorithms
<http://www.bjarni-gunnarsson.net>

Tasks

Task 1, CDP

Use the *hita.wav* (or b or c) sound provided with the course as base material. The sounds are short and we will need to transform it to focus on its different properties. The goal is to create different version of the sound with **CDP** and play these back using **SuperCollider**. The following two operations should be applied at least more than once:

1. Extend the sound in duration (see CDP extend).
2. Distort the sound with wavecycle-based processes (see CDP distort).

Applying these operations several time will result in **different paths** from source to destination. The final step is to implement **sample-playback strategies** with SuperCollider to navigate the space of the

Task 2, CDP

Use any of the sounds from the **abstract** or **concrete** folders.

Systematically create hybrid sounds of both sources using cross-synthesis such as:

- * Formants impose
 - * Spectral vocode
 - * Envelope follow
 - * Spectral Diff
 - * Morph
- Etc.

Once the hybrid sounds are ready implement a strategy to grow through them.

Task 3, CDP

Use any of the sounds from the **abstract** or **concrete** folders.

Explore **batch processing**.

Choose a set of at least 4 different CDP processes and then use the CDPbatch class to execute them.

Batch Processing Usage

```
// Process all .wav files in a directory with random parameters
~dir = "/path/to/audio/files/";

// Easiest method - just specify parameter ranges:
CDPBatch.ranges(~dir, 1, [
  ["distort average", [12, 80], "_average"],
  ["distort overload 2", [[0.001, 0.025], [0.8, 0.99], [100, 4000]], "_overload"],
  ["blur scatter", [[3, 12], "4"], "_scatter"]
]);

// Multiple passes - reprocess outputs recursively:
CDPBatch.ranges(~dir, 1, [
  ["distort average", [12, 80], "_average"]
], passes: 3); // Processes 3 times: original -> pass1 -> pass2 -> pass3

// Or use custom parameter functions:
CDPBatch.custom(~dir, 1, [
  ["distort average", {|b| b.rrange(12, 80).asString}, "_average"],
  ["distort repeat", {|b| b.rrange(2, 6).asString + " -c40"}, "_repeat"]
]);

// Or use preset batch classes:
CDPBatchDistort.process(~dir, 1); // Creates 8 variations per file
CDPBatchCycles.process(~dir, 2); // Cycle-based transformations
CDPBatchExtend.process(~dir, 1); // Extension processes
CDPBatchMulti.process(~dir, 1); // Complex multi-layered process

// Output appears in: /path/to/audio/files/batch_output/
```

See `BATCH_README.md` and `CDP.Batch.scd` for complete batch processing documentation and examples.

